디지털 논리

Quine-McCluskey Algorithm

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Objectives

1. **Find the prime implicants of a function by using Quine-McCluskey method.**
   - ⇒ Systematic simplification procedure for programmable coding
   - ⇒ Applicable for more than 5 variables (cf: Karnaugh map)
2. **Determine the essential prime implicants using prime implicant chart.**
3. **Find a minimum sum-of-products using Petrick's method.**
4. **Minimize an incompletely specified function (with don't care's) using Quine-McCluskey method.**

Combining Minterms

Two terms can be combined if they differ in exactly one variable.

=> 2 variables: XY+XY'=X

=> 3 variables: XYZ + XYZ'=XY

All possible pairs of minterms should be compared.

- The binary notation and its algebraic equivalent form

$$AB'CD' + AB'CD = AB'C$$

$$\underbrace{1\ 0\ 1}_{X}\ \underset{Y}{0} + \underbrace{1\ 0\ 1}_{X}\ \underset{Y'}{1} = \underbrace{1\ 0\ 1}_{X}\ -$$

=> the dash indicates a missing variable

Determining Prime Implicants (I)

The binary minterms are sorted into groups according to # of '1' in each minterm in order to reduce the comparison.

$$f(a,b,c,d) = \sum m(0,1,2,5,6,7,8,9,10,14)$$

group 0	0	0000
	1	0001
group 1	2	0010
	8	1000
	5	0101
group 2	6	0110
	9	1001
	10	1010
group 3	7	0111
	14	1110

Only terms in adjacent groups must be compared.

Determining Prime Implicants (II)

	Column I		Column II		Column III	
group 0	0	0000 ✓	0, 1	000- ✓	0, 1, 8, 9	-00-
group 1	1	0001 ✓	0, 2	00-0 ✓	0, 2, 8, 10	-0-0
	2	0010 ✓	0, 8	-000 ✓	~~0, 8, 1, 9~~	~~-00-~~
	8	1000 ✓	1, 5	0-01	~~0, 2, 8, 10~~	~~-0-0~~
group 2	5	0101 ✓	1, 9	-001 ✓	2, 6, 10, 14	--10
	6	0110 ✓	2, 6	0-10 ✓	~~2, 10, 6, 14~~	~~--10~~
	9	1001 ✓	2, 10	-010 ✓		
	10	1010 ✓	8, 9	100- ✓		
group 3	7	0111 ✓	8, 10	10-0 ✓		
	14	1110 ✓	5, 7	01-1		
			6, 7	011-		
			6, 14	-110 ✓		
			10, 14	1-10 ✓		

Duplicates are deleted

Prime implicant: No check (✓)

Eliminating redundant terms

The function is equal to the sum of prime implicants

$$f = \underset{(1,5)}{a'c'd} + \underset{(5,7)}{a'bd} + \underset{(6,7)}{a'bc} + \underset{(0,1,8,9)}{b'c'} + \underset{(0,2,8,10)}{b'd'} + \underset{(2,6,10,14)}{cd'}$$

$a'c'd$

$b'd'$

$a'bc$

Using the consensus theorem

$$f = a'bd + b'c' + cd'$$

Any systematic way to find essential and remaining P.I ?

Prime Implicant Chart (I)

A systematic way to find the (essential) prime implicants

Essential Prime Implicant : ⊗

		0	1	2	5	6	7	8	9	10	14
(0, 1, 8, 9)	$b'c'$	x	x					x	⊗		
(0, 2, 8, 10)	$b'd'$	x		x				x		x	
(2, 6, 10, 14)	cd'			x		x				x	⊗
(1, 5)	$a'c'd$		x		x						
(5, 7)	$a'bd$				x		x				
(6, 7)	$a'bc$					x	x				

Essential prime implicant includes the minterm which appears only once in the chart.

Prime Implicant Chart (II)

Essential PI should be first selected for expansion, then select minimal PI's for remaining minterms.

		0	1	2	5	6	7	8	9	10	14
$(0,1,8,9)$	$b'c'$	x	x					x	x		
$(0,2,8,10)$	$b'd'$	x		x				x		x	
$(2,6,10,14)$	cd'			x		x				x	x
$(1,5)$	$a'c'd$		x		x						
$(5,7)$	$a'bd$				x		x				
$(6,7)$	$a'bc$					x	x				

$$f = b'c' + cd' + a'bd$$

We need another method to select PI for remaining minterms systematically.

Ex: Prime implicant chart (I)

There are multiple minimum sum of products.

$$F(a,b,c) = \sum m(0,1,2,5,6,7)$$

0	000	✓	0,1	00-	$a'b'$
1	001	✓	0,2	0-0	$a'c'$
2	010	✓	1,5	-01	$b'c$
5	101	✓	2,6	-10	bc'
6	110	✓	5,7	1-1	ac
7	111	✓	6,7	11-	ab

Ex: Prime implicant chart (II)

The resulting prime implicant chart

			0	1	2	5	6	7
①	→ (0, 1)	$a'b'$	x	x				
	(0,2)	$a'c'$	x		x			
	(1,5)	$b'c$		x		x		
②	→(2,6)	bc'			x		x	
③	→(5,7)	ac				x		x
	(6,7)	ab					x	x

One solution:

$$F = a'b' + bc' + ac$$

Ex: Prime implicant chart (III)

Again starting with the other prime implicant that covers column 0.

				0	1	2	5	6	7
	P_1	(0,1)	$a'b'$	x	x				
①	P_2	(0,2)	$a'c'$	x		x			
②	P_3	(1,5)	$b'c$		x		x		
	P_4	(2,6)	bc'			x		x	
	P_5	(5,7)	ac				x		x
③	P_6	(6,7)	ab					x	x

Another minimum solution：

$$F = a'c' + b'c + ab$$

Petrick's Method (I)

A technique for determining all minimum sum of products from a PI chart

label			0	1	2	5	6	7
P_1	(0, 1)	$a'b'$	x	x				
P_2	(0,2)	$a'c'$	x		x			
P_3	(1,5)	$b'c$		x		x		
P_4	(2,6)	bc'			x		x	
P_5	(5,7)	ac				x		x
P_6	(6,7)	ab					x	x

Because we must cover all the minterms, following function P must be true:

$$P = (P_1 + P_2)(P_1 + P_3)(P_2 + P_4)(P_3 + P_5)(P_4 + P_6)(P_5 + P_6) = 1$$

minterm0 minterm1 ..... minterm7

Petrick's Method (II)

Reduce P to a minimum sum of product

First, we multiply out, using (X+Y)(X+Z) = X+YZ Distributive law

$$
\begin{aligned}
P &= (P_1 + P_2P_3)(P_4 + P_2P_6)(P_5 + P_3P_6) \\
&= (P_1P_4 + P_1P_2P_6 + P_2P_3P_4 + P_2P_3P_6)(P_5 + P_3P_6) \\
&= P_1P_4P_5 + P_1P_2P_5P_6 + P_2P_3P_4P_5 + \cancel{P_2P_3P_5P_6} + P_1P_3P_4P_6 \\
&\quad + \cancel{P_1P_2P_3P_6} + \cancel{P_2P_3P_4P_6} + \underline{P_2P_3P_6}
\end{aligned}
$$

P_1	(0, 1)	$a'b'$
P_2	(0,2)	$a'c'$
P_3	(1,5)	$b'c$
P_4	(2,6)	bc'
P_5	(5,7)	ac
P_6	(6,7)	ab

Use X+XY=X to eliminate redundant terms from P

$$P = P_1P_4P_5 + P_1P_2P_5P_6 + P_2P_3P_4P_5 + P_1P_3P_4P_6 + P_2P_3P_6$$

Choose *P1,P4,P5* or *P2,P3,P6* for minimum solution

$$F = a'b' + bc' + ac$$ or $$F = a'c' + b'c + ab$$

Incomplete Function (I)

When we have the don't care (X) for some circuits, the don't cares are included in the first column.

$$F(A,B,C,D) = \sum m(2,3,7,9,11,13) + \sum d(1,10,15)$$

1	0001 ✓	(1, 3)	00-1 ✓	(1, 3, 9, 11)	-0-1	$B'D$
2	0010 ✓	(1, 9)	-001 ✓	(2, 3, 10, 11)	-01-	$B'C$
3	0011 ✓	(2, 3)	001- ✓	(3, 7, 11, 15)	--11	CD
9	1001 ✓	(2, 10)	-010 ✓	(9, 11, 13, 15)	1--1	AD
10	1010 ✓	(3, 7)	0-11 ✓			
7	0111 ✓	(3, 11)	-011 ✓			
11	1011 ✓	(9, 11)	10-1 ✓			
13	1101 ✓	(9, 13)	1-01 ✓			
15	1111 ✓	(10, 11)	101- ✓			
		(7, 15)	-111 ✓			
		(11, 15)	1-11 ✓			
		(13, 15)	11-1 ✓			

Incomplete Function (II)

Don't care's are omitted when forming the PI chart.

$$F(A,B,C,D) = \sum m(2,3,7,9,11,13) + \sum d(1,10,15)$$

		2	3	7	9	11	13
$B'D$	$(1,3,9,11)$		x		x	x	
$B'C$	$*\ (2,3,10,11)$	x	x			x	
CD	$*\ (3,7,11,15)$		x	x		x	
AD	$*\ (9,11,13,15)$				x	x	x

$$F = B'C + CD + AD$$

=> don't care terms d(10,15)는 1로 d(1)은 0으로 설정